

31. BRANCHES AND DERIVATIVES OF LOGARITHMS

If $z = re^{i\theta}$ is a nonzero complex number, the argument θ has any one of the values $\theta = \Theta + 2n\pi$ ($n = 0, \pm 1, \pm 2, \dots$), where $\Theta = \text{Arg } z$. Hence the definition

$$\log z = \ln r + i(\Theta + 2n\pi) \quad (n = 0, \pm 1, \pm 2, \dots)$$

of the multiple-valued logarithmic function in Sec. 30 can be written

$$(1) \quad \log z = \ln r + i\theta.$$

If we let α denote any real number and restrict the value of θ in expression (1) so that $\alpha < \theta < \alpha + 2\pi$, the function

$$(2) \quad \log z = \ln r + i\theta \quad (r > 0, \alpha < \theta < \alpha + 2\pi),$$

with components

$$(3) \quad u(r, \theta) = \ln r \quad \text{and} \quad v(r, \theta) = \theta,$$

is *single-valued* and continuous in the stated domain (Fig. 35). Note that if the function (2) were to be defined on the ray $\theta = \alpha$, it would not be continuous there. For if z is a point on that ray, there are points arbitrarily close to z at which the values of v are near α and also points such that the values of v are near $\alpha + 2\pi$.

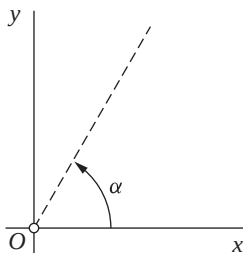


FIGURE 35

The function (2) is not only continuous but also analytic throughout the domain $r > 0, \alpha < \theta < \alpha + 2\pi$ since the first-order partial derivatives of u and v are continuous there and satisfy the polar form (Sec. 23)

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

of the Cauchy–Riemann equations. Furthermore, according to Sec. 23,

$$\frac{d}{dz} \log z = e^{-i\theta} (u_r + iv_r) = e^{-i\theta} \left(\frac{1}{r} + i0 \right) = \frac{1}{re^{i\theta}};$$

that is,

$$(4) \quad \frac{d}{dz} \log z = \frac{1}{z} \quad (|z| > 0, \alpha < \arg z < \alpha + 2\pi).$$

In particular,

$$(5) \quad \frac{d}{dz} \operatorname{Log} z = \frac{1}{z} \quad (|z| > 0, -\pi < \operatorname{Arg} z < \pi).$$

A *branch* of a multiple-valued function f is any single-valued function F that is analytic in some domain at each point z of which the value $F(z)$ is one of the values of f . The requirement of analyticity, of course, prevents F from taking on a random selection of the values of f . Observe that for each fixed α , the single-valued function (2) is a branch of the multiple-valued function (1). The function

$$(6) \quad \operatorname{Log} z = \ln r + i\Theta \quad (r > 0, -\pi < \Theta < \pi)$$

is called the *principal branch*.

A *branch cut* is a portion of a line or curve that is introduced in order to define a branch F of a multiple-valued function f . Points on the branch cut for F are singular points (Sec. 24) of F , and any point that is common to all branch cuts of f is called a *branch point*. The origin and the ray $\theta = \alpha$ make up the branch cut for the branch (2) of the logarithmic function. The branch cut for the principal branch (6) consists of the origin and the ray $\Theta = \pi$. The origin is evidently a branch point for branches of the multiple-valued logarithmic function.

Special care must be taken in using branches of the logarithmic function, especially since expected identities involving logarithms do not always carry over from calculus.

EXAMPLE. When the principal branch (6) is used, one can see that

$$\operatorname{Log}(i^3) = \operatorname{Log}(-i) = \ln 1 - i\frac{\pi}{2} = -\frac{\pi}{2}i$$

and

$$3 \operatorname{Log} i = 3 \left(\ln 1 + i\frac{\pi}{2} \right) = \frac{3\pi}{2}i.$$

Hence

$$\operatorname{Log}(i^3) \neq 3 \operatorname{Log} i.$$

(See also Exercises 3 and 4.)

In Sec. 32, we shall derive some identities involving logarithms that *do* carry over from calculus, sometimes with qualifications as to how they are to be interpreted. A reader who wishes to pass to Sec. 33 can simply refer to results in Sec. 32 when needed.

EXERCISES

1. Show that

$$(a) \operatorname{Log}(-ei) = 1 - \frac{\pi}{2}i; \quad (b) \operatorname{Log}(1-i) = \frac{1}{2} \ln 2 - \frac{\pi}{4}i.$$

2. Show that

$$(a) \log e = 1 + 2n\pi i \quad (n = 0, \pm 1, \pm 2, \dots);$$

$$(b) \log i = \left(2n + \frac{1}{2}\right)\pi i \quad (n = 0, \pm 1, \pm 2, \dots);$$

$$(c) \log(-1 + \sqrt{3}i) = \ln 2 + 2\left(n + \frac{1}{3}\right)\pi i \quad (n = 0, \pm 1, \pm 2, \dots).$$

3. Show that

$$(a) \operatorname{Log}(1+i)^2 = 2 \operatorname{Log}(1+i); \quad (b) \operatorname{Log}(-1+i)^2 \neq 2 \operatorname{Log}(-1+i).$$

4. Show that

$$(a) \log(i^2) = 2 \log i \quad \text{when} \quad \log z = \ln r + i\theta \left(r > 0, \frac{\pi}{4} < \theta < \frac{9\pi}{4}\right);$$

$$(b) \log(i^2) \neq 2 \log i \quad \text{when} \quad \log z = \ln r + i\theta \left(r > 0, \frac{3\pi}{4} < \theta < \frac{11\pi}{4}\right).$$

5. Show that

$$(a) \text{ the set of values of } \log(i^{1/2}) \text{ is}$$

$$\left(n + \frac{1}{4}\right)\pi i \quad (n = 0, \pm 1, \pm 2, \dots)$$

and that the same is true of $(1/2) \log i$;

$$(b) \text{ the set of values of } \log(i^2) \text{ is not the same as the set of values of } 2 \log i.$$

6. Given that the branch $\log z = \ln r + i\theta$ ($r > 0, \alpha < \theta < \alpha + 2\pi$) of the logarithmic function is analytic at each point z in the stated domain, obtain its derivative by differentiating each side of the identity (Sec. 30)

$$e^{\log z} = z \quad (z \neq 0)$$

and using the chain rule.

7. Find all roots of the equation $\log z = i\pi/2$.

$$\text{Ans. } z = i.$$

8. Suppose that the point $z = x + iy$ lies in the horizontal strip $\alpha < y < \alpha + 2\pi$. Show that when the branch $\log z = \ln r + i\theta$ ($r > 0, \alpha < \theta < \alpha + 2\pi$) of the logarithmic function is used, $\log(e^z) = z$. [Compare with equation (4), Sec. 30.]

9. Show that

$$(a) \text{ the function } f(z) = \operatorname{Log}(z-i) \text{ is analytic everywhere except on the portion } x \leq 0 \text{ of the line } y = 1;$$

$$(b) \text{ the function}$$

$$f(z) = \frac{\operatorname{Log}(z+4)}{z^2+i}$$

is analytic everywhere except at the points $\pm(1-i)/\sqrt{2}$ and on the portion $x \leq -4$ of the real axis.

10. Show in two ways that the function $\ln(x^2 + y^2)$ is harmonic in every domain that does not contain the origin.

11. Show that

$$\operatorname{Re}[\log(z-1)] = \frac{1}{2} \ln[(x-1)^2 + y^2] \quad (z \neq 1).$$

Why must this function satisfy Laplace's equation when $z \neq 1$?

32. SOME IDENTITIES INVOLVING LOGARITHMS

If z_1 and z_2 denote any two nonzero complex numbers, it is straightforward to show that

$$(1) \quad \log(z_1 z_2) = \log z_1 + \log z_2.$$

This statement, involving a multiple-valued function, is to be interpreted in the same way that the statement

$$(2) \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2$$

was in Sec. 8. That is, if values of two of the three logarithms are specified, then there is a value of the third such that equation (1) holds.

The verification of statement (1) can be based on statement (2) in the following way. Since $|z_1 z_2| = |z_1| |z_2|$ and since these moduli are all positive real numbers, we know from experience with logarithms of such numbers in calculus that

$$\ln |z_1 z_2| = \ln |z_1| + \ln |z_2|.$$

So it follows from this and equation (2) that

$$(3) \quad \ln |z_1 z_2| + i \arg(z_1 z_2) = (\ln |z_1| + i \arg z_1) + (\ln |z_2| + i \arg z_2).$$

Finally, because of the way in which equations (1) and (2) are to be interpreted, equation (3) is the same as equation (1).

EXAMPLE. To illustrate statement (1), write $z_1 = z_2 = -1$ and recall from Examples 2 and 3 in Sec. 30 that

$$\log 1 = 2n\pi i \quad \text{and} \quad \log(-1) = (2n+1)\pi i,$$

where $n = 0, \pm 1, \pm 2, \dots$. Noting that $z_1 z_2 = 1$ and using the values

$$\log(z_1 z_2) = 0 \quad \text{and} \quad \log z_1 = \pi i,$$

we find that equations (1) is satisfied when the value $\log z_2 = -\pi i$ is chosen.

If, on the other hand, the principal values

$$\operatorname{Log} 1 = 0 \quad \text{and} \quad \operatorname{Log}(-1) = \pi i$$

are used,

$$\operatorname{Log}(z_1 z_2) = 0 \quad \text{and} \quad \operatorname{Log} z_1 + \log z_2 = 2\pi i$$

for the same numbers z_1 and z_2 . Thus statement (1), which is sometimes true when $\log$ is replaced by Log (see Exercise 1), is not always true when principal values are used in all three of its terms.

Verification of the statement

$$(4) \quad \log\left(\frac{z_1}{z_2}\right) = \log z_1 - \log z_2,$$

which is to be interpreted in the same way as statement (1), is left to the exercises.

We include here two other properties of $\log z$ that will be of special interest in Sec. 33. If z is a nonzero complex number, then

$$(5) \quad z^n = e^{n \log z} \quad (n = 0 \pm 1, \pm 2, \dots)$$

for any value of $\log z$ that is taken. When $n = 1$, this reduces, of course, to relation (3), Sec. 30. Equation (5) is readily verified by writing $z = re^{i\theta}$ and noting that each side becomes $r^n e^{in\theta}$.

It is also true that when $z \neq 0$,

$$(6) \quad z^{1/n} = \exp\left(\frac{1}{n} \log z\right) \quad (n = 1, 2, \dots).$$

That is, the term on the right here has n distinct values, and those values are the n th roots of z . To prove this, we write $z = r \exp(i\Theta)$, where Θ is the principal value of $\arg z$. Then, in view of definition (2), Sec. 30, of $\log z$,

$$\exp\left(\frac{1}{n} \log z\right) = \exp\left[\frac{1}{n} \ln r + \frac{i(\Theta + 2k\pi)}{n}\right]$$

where $k = 0, \pm 1, \pm 2, \dots$. Thus

$$(7) \quad \exp\left(\frac{1}{n} \log z\right) = \sqrt[n]{r} \exp\left[i\left(\frac{\Theta}{n} + \frac{2k\pi}{n}\right)\right] \quad (k = 0, \pm 1, \pm 2, \dots).$$

Because $\exp(i2k\pi/n)$ has distinct values only when $k = 0, 1, \dots, n-1$, the right-hand side of equation (7) has only n values. That right-hand side is, in fact, an expression for the n th roots of z (Sec. 9), and so it can be written $z^{1/n}$. This establishes property (6), which is actually valid when n is a negative integer too (see Exercise 5).